University of Technical Education HCMC
Faculty of Information Technology
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SOFTWARE DEVELOPMENT ENVIRONMENTS

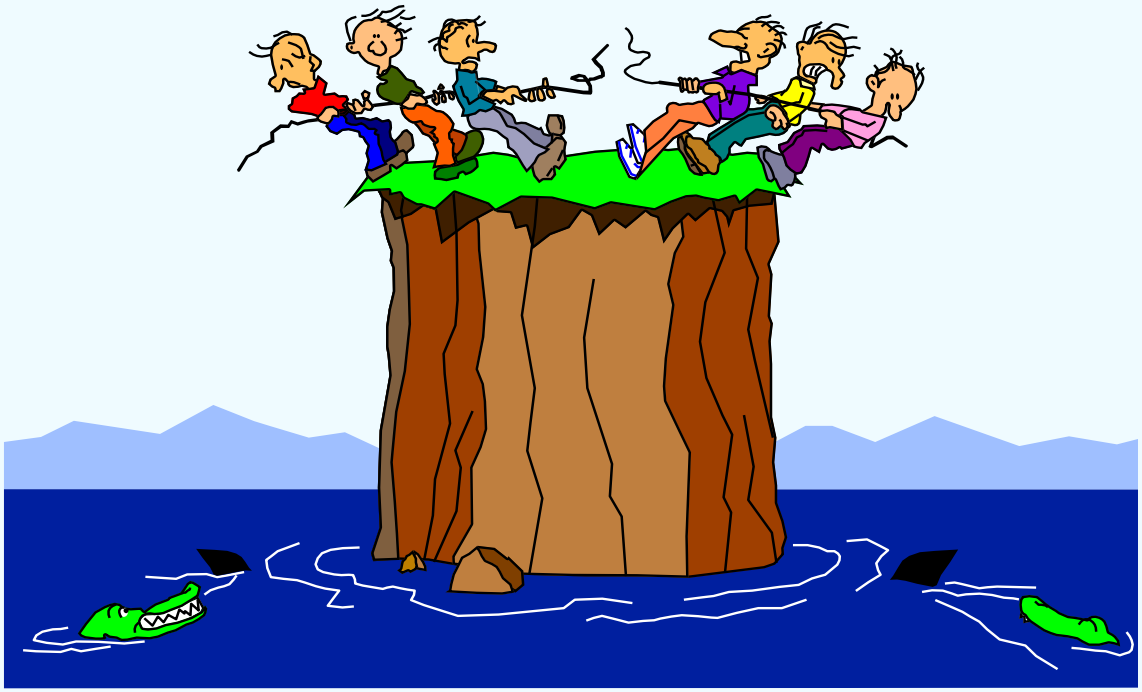
Chapter 02

Unified Modeling Language - UML

Lecturer: Nguyễn Trần Thi Văn

Lesson 04

Interaction Diagrams



Sequence Diagrams

What are Interaction Diagrams?

- An interaction diagram is a graphical representation of interactions between objects
- There are two kinds of interaction diagrams
  - Sequence diagrams
  - Collaboration (communication) diagrams
- Each provides a different view of the same interaction
  - Sequence diagrams are time ordered
  - Collaboration diagrams may include data flow



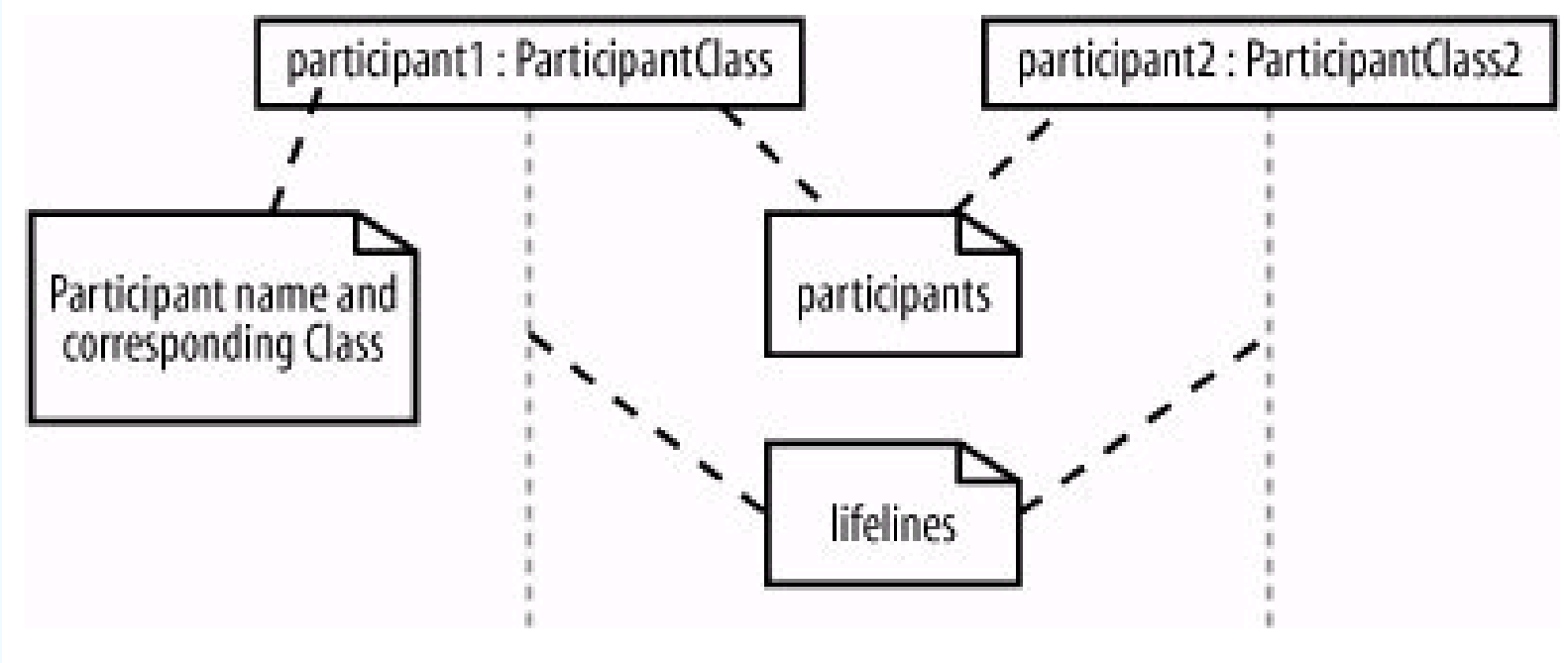
What is a Sequence Diagram?

- A sequence diagram shows object interactions arranged in time sequence
- The diagram shows
  - The objects participating in the interaction
  - The sequence of messages exchanged
- A sequence diagram contains:
  - Objects with their "lifelines"
  - Messages exchanged between objects in ordered sequence
  - Focus of control (optional)



Participants

- **Participants in sequence diagrams are objects**



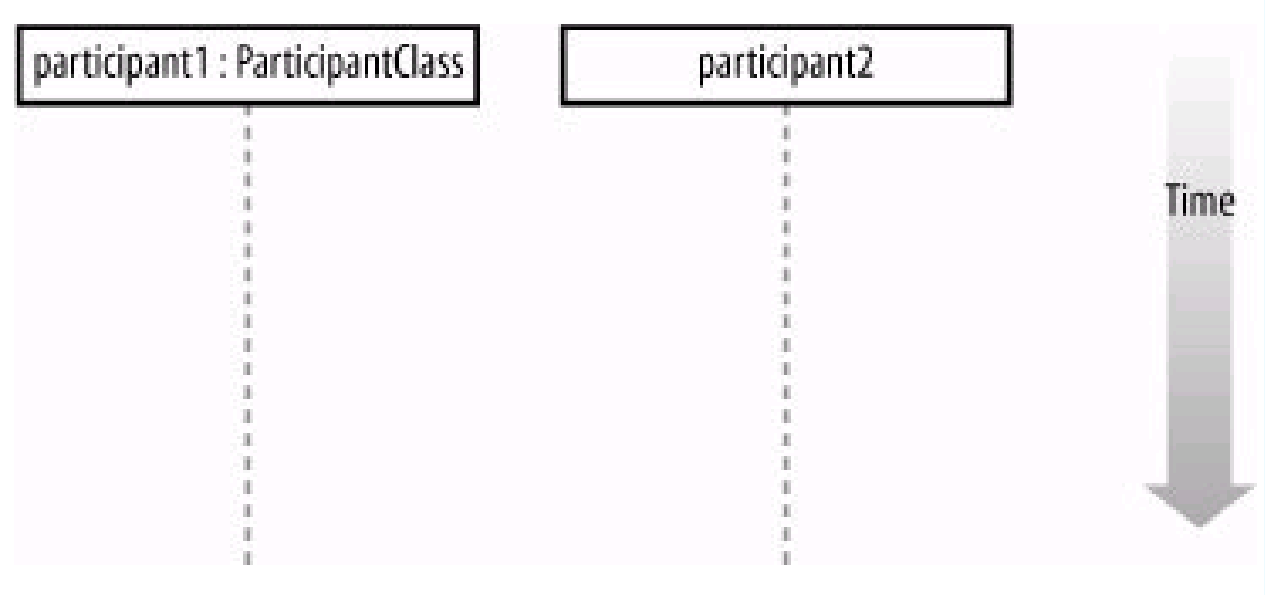
Naming Objects in Sequence Diagrams

admin	Object name, not assigned to a specific class
:BlogEntry	Object without a name, class name only
admin:Administrator	Object name with class
b [10]:BlogEntry	Array of 10 objects
:CMS ref cmsInteraction	Reference to another sequence diagram

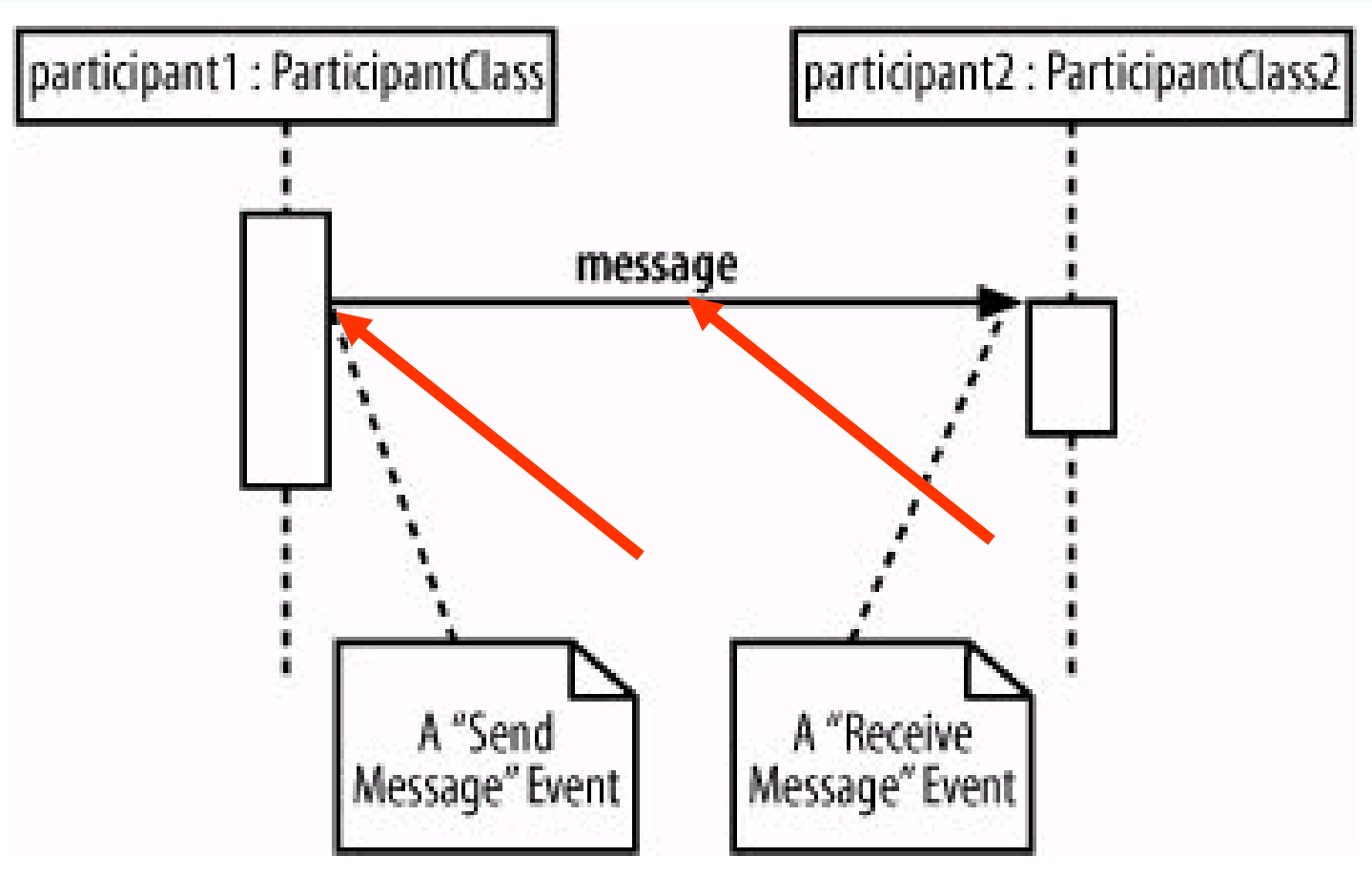


Time

- **Time runs down from top of the diagram**

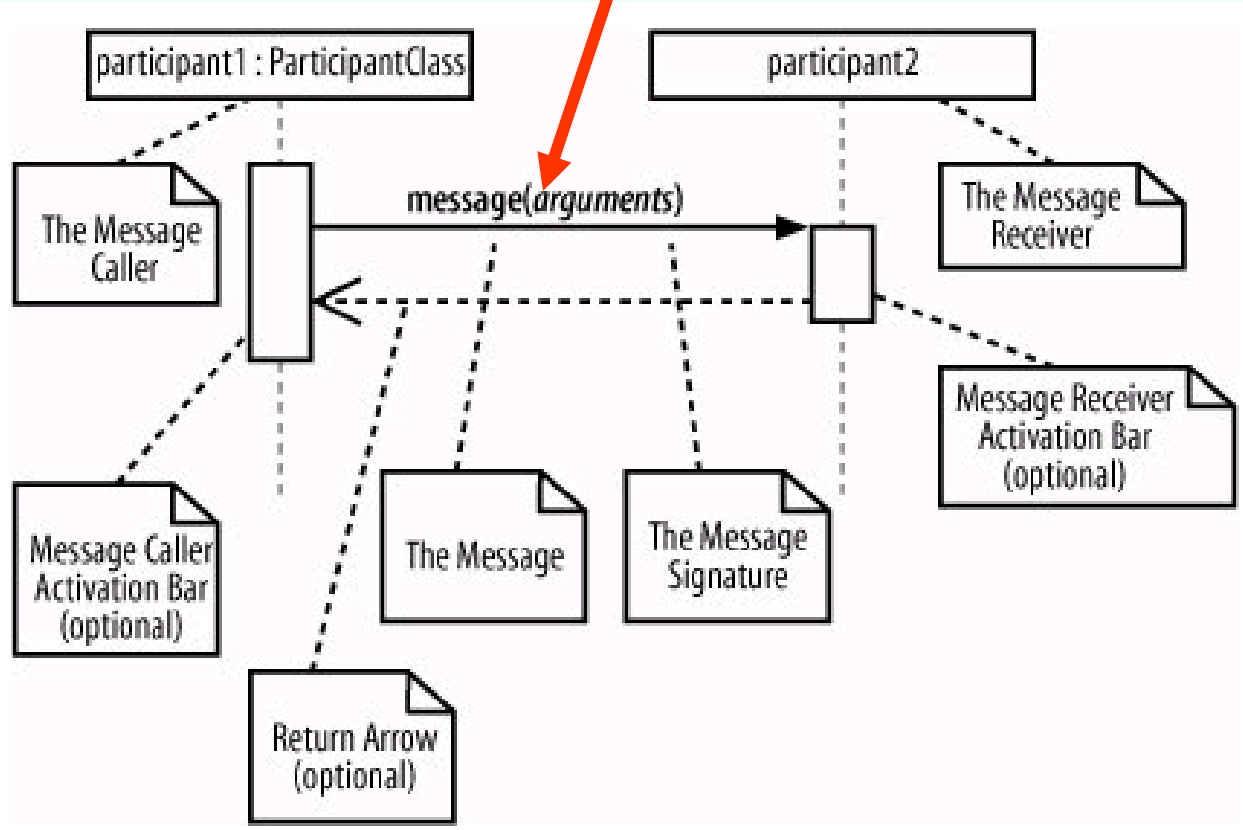


Events and Messages



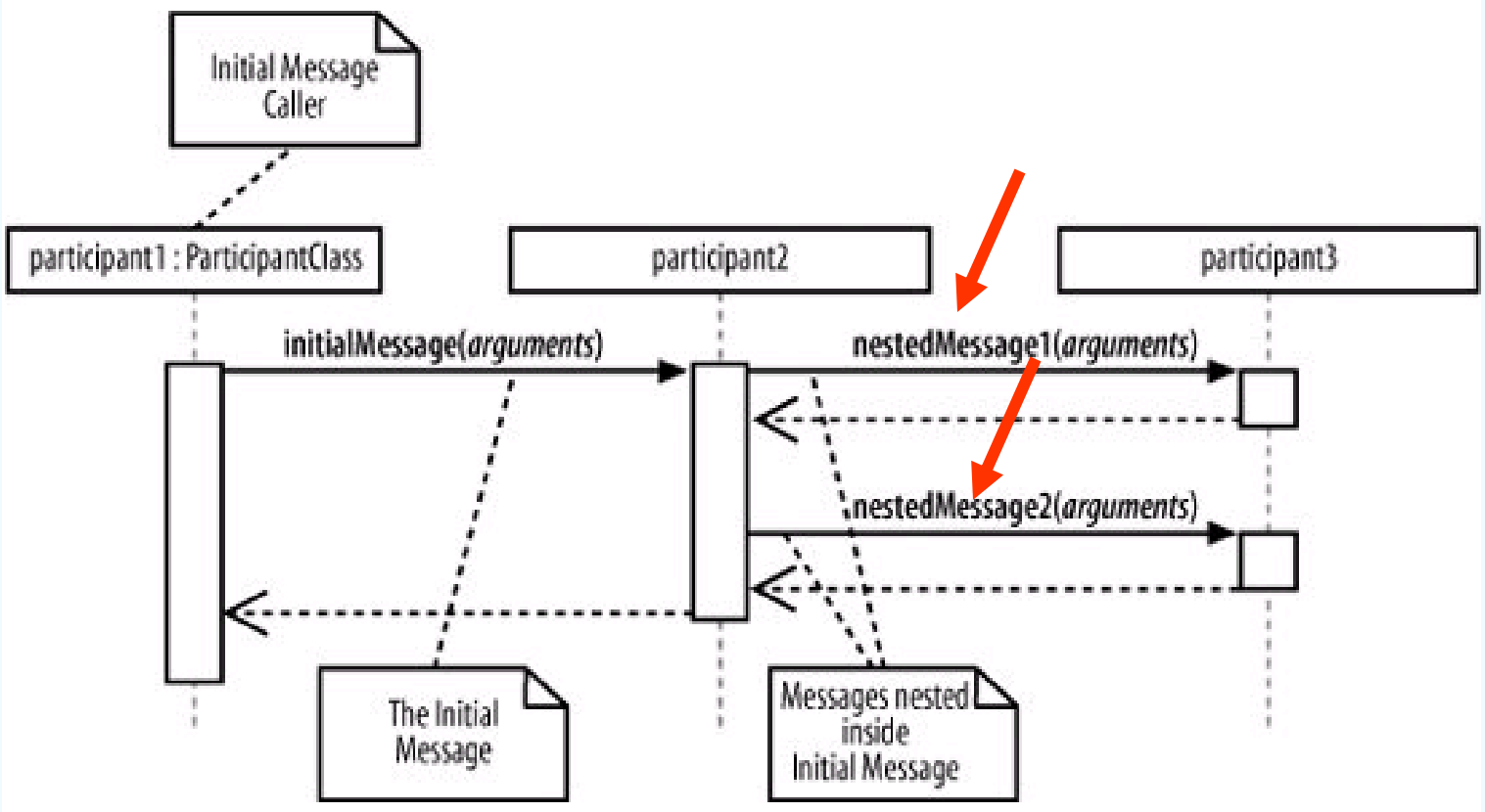
Events and Messages (2)

- **attribute=message_name(arguments):return_type**



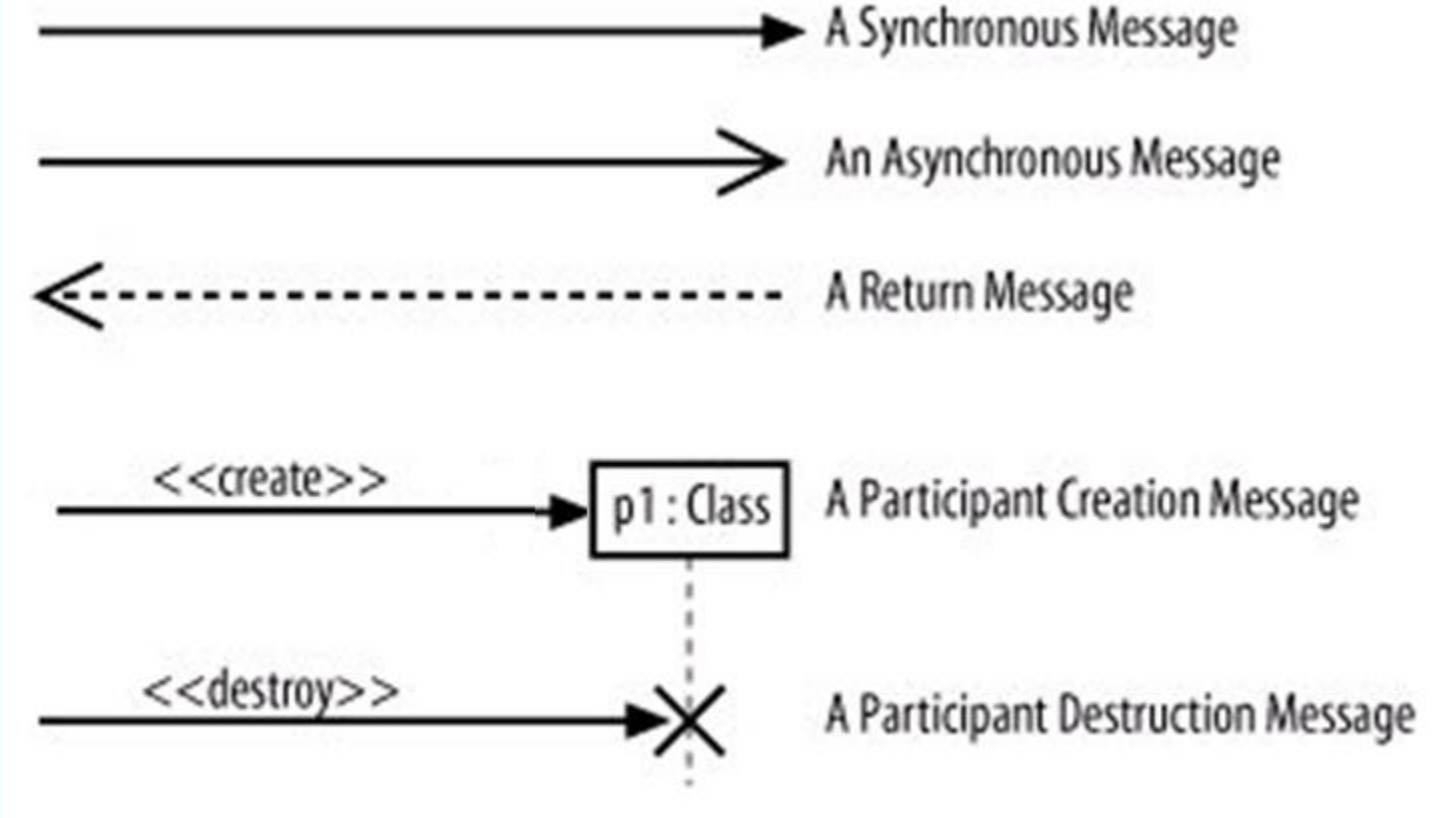
Nested messages

- 2 nested messages are invoked when an initial message is recieved



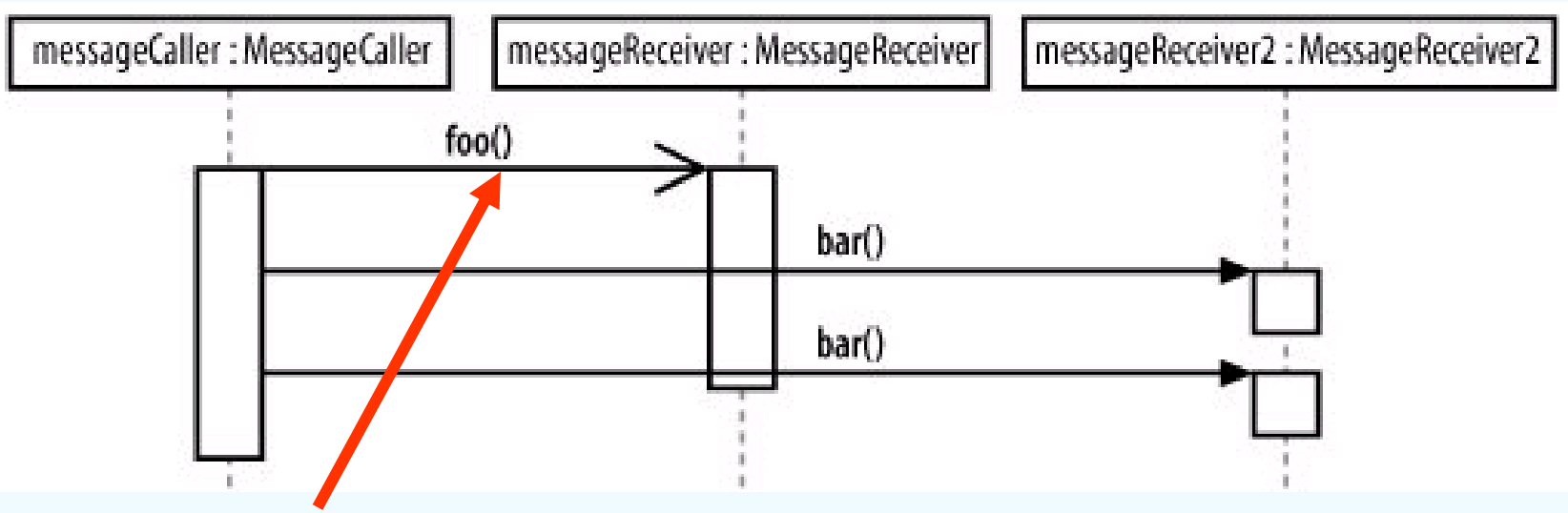
Message arrows

- **There are 5 types of message arrows:**

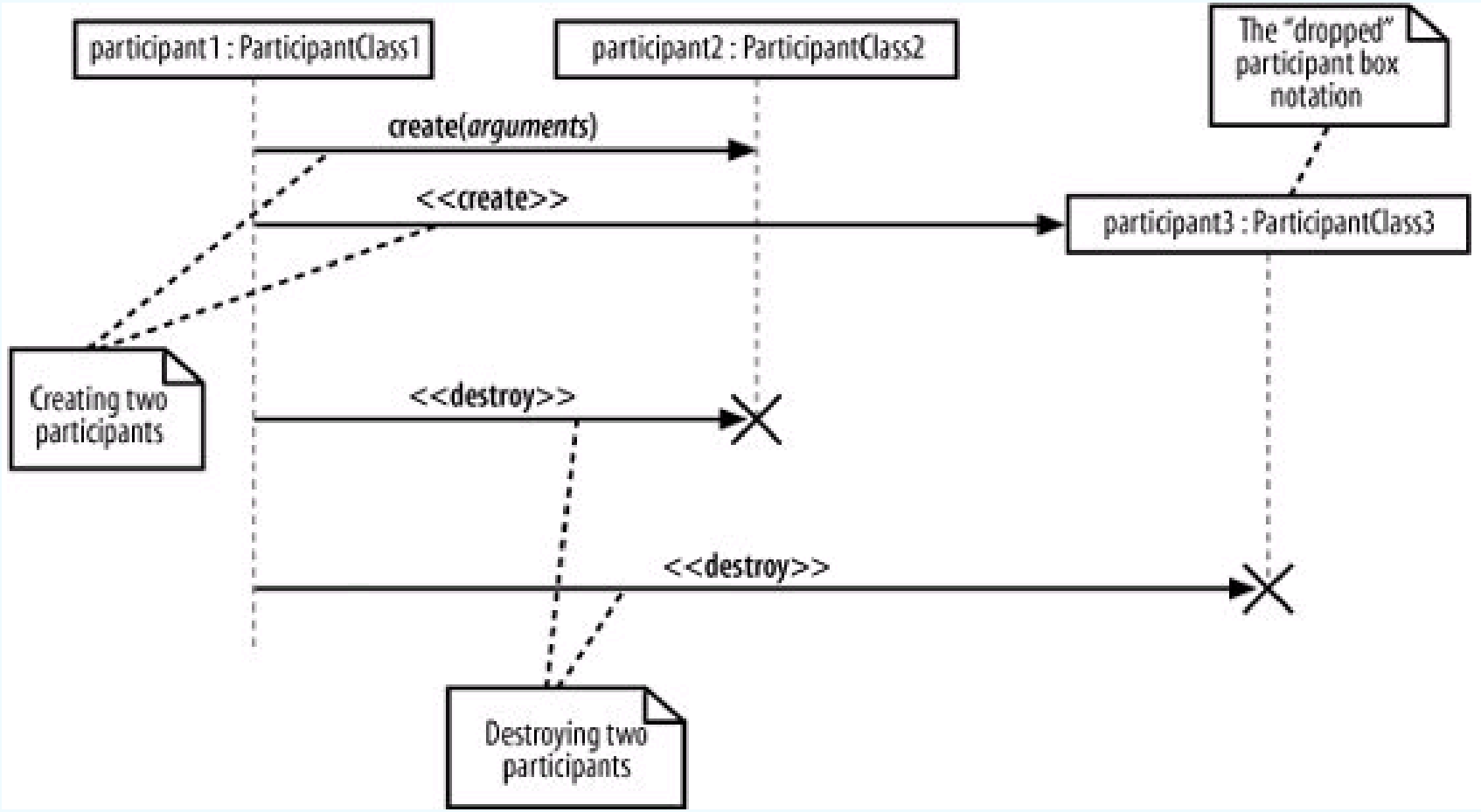


Message arrows - Async

- **Example:**

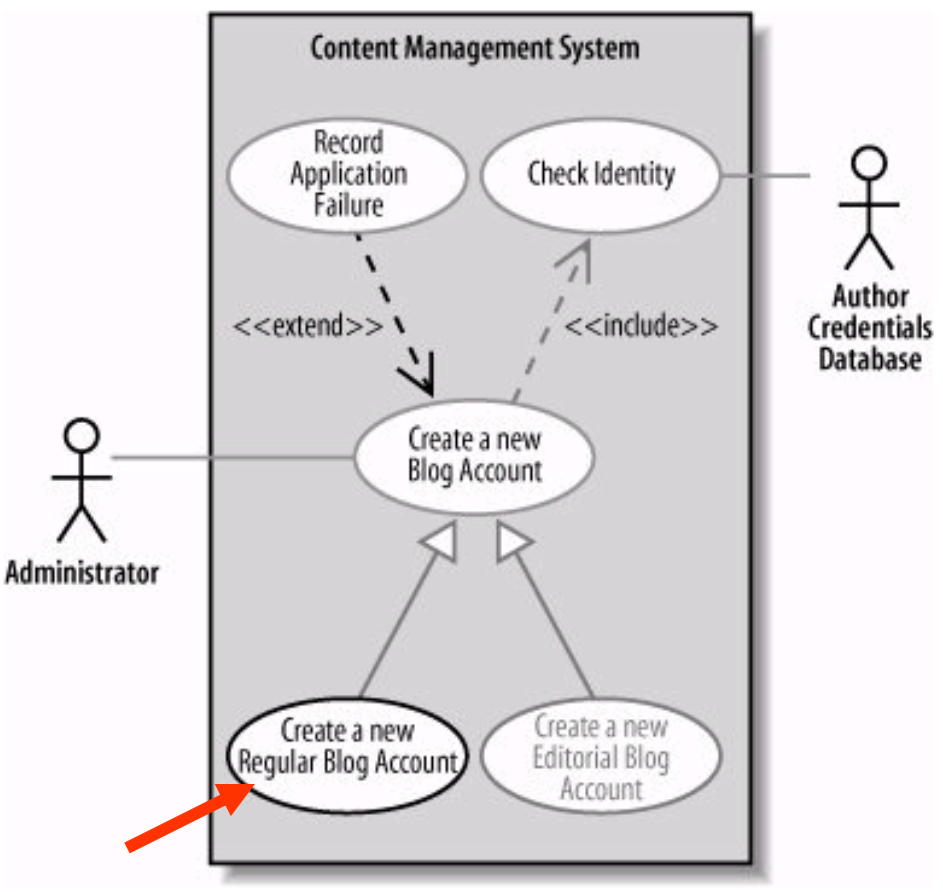


<<create>> and <<destroy>>



Realization of use case

- Each use case may have 1 or more sequence diagram
- Sequence diagram describe the flows of events



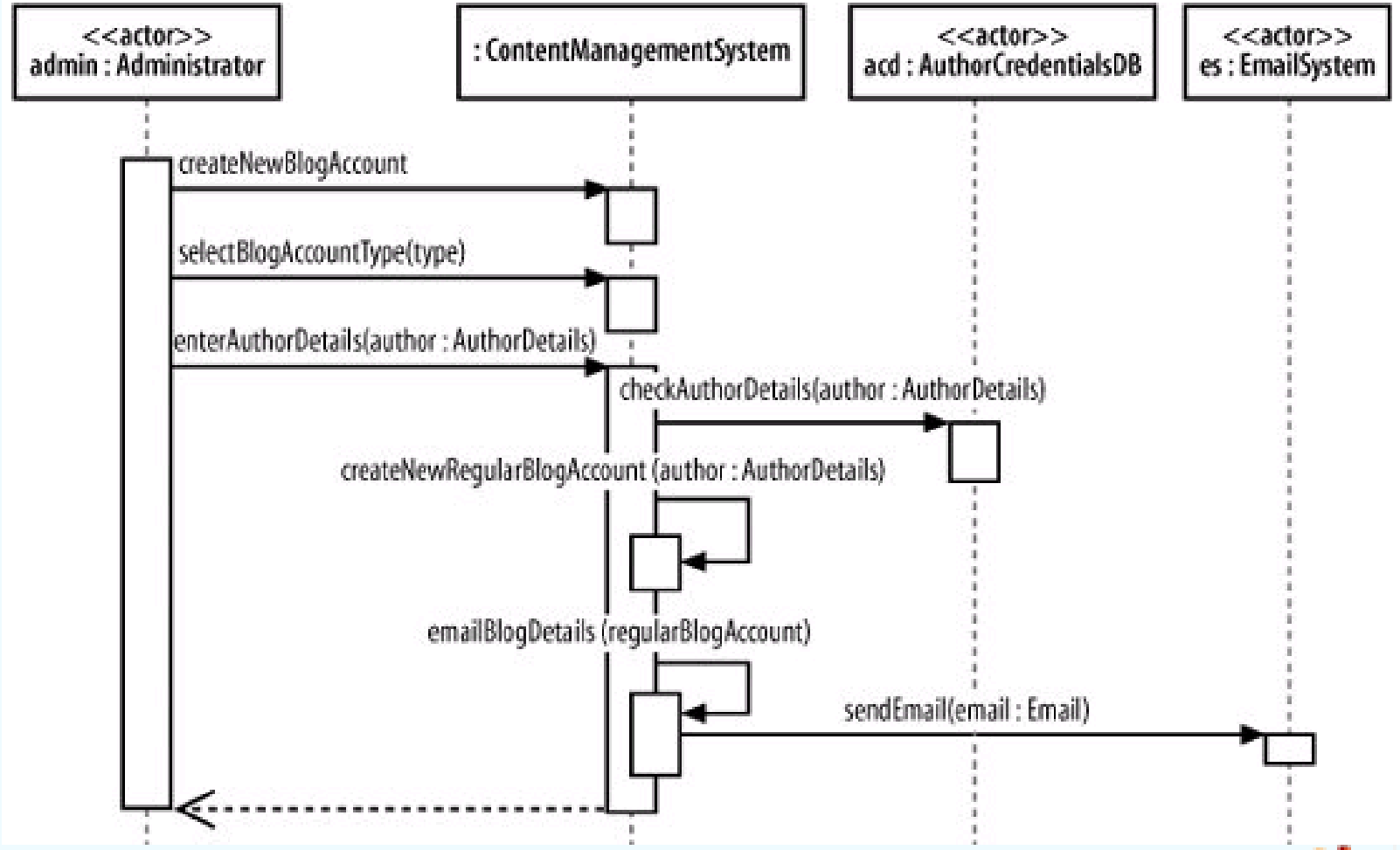
Realization of use case (2)

- **Create new Regular account use case**

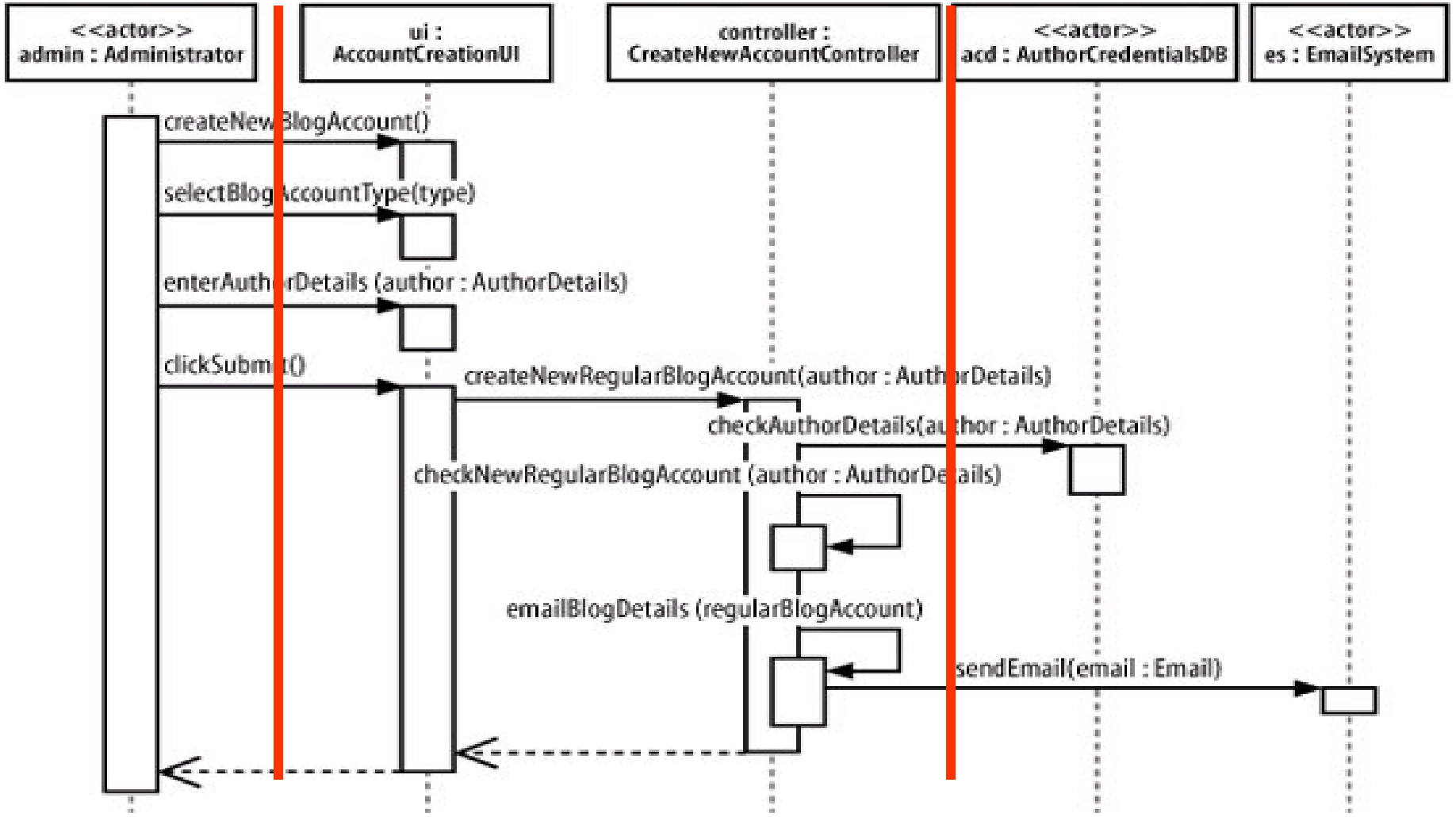
Main Flow	Step	Action
	1	The Administrator asks the system to create a new blog account.
	2	The Administrator selects the regular blog account type.
	3	The Administrator enters the author's details.
	4	The author's details are checked using the Author Credentials Database.
	5	The new regular blog account is created.
	6	A summary of the new blog account's details are emailed to the author.



Top level sequence diagram



More details

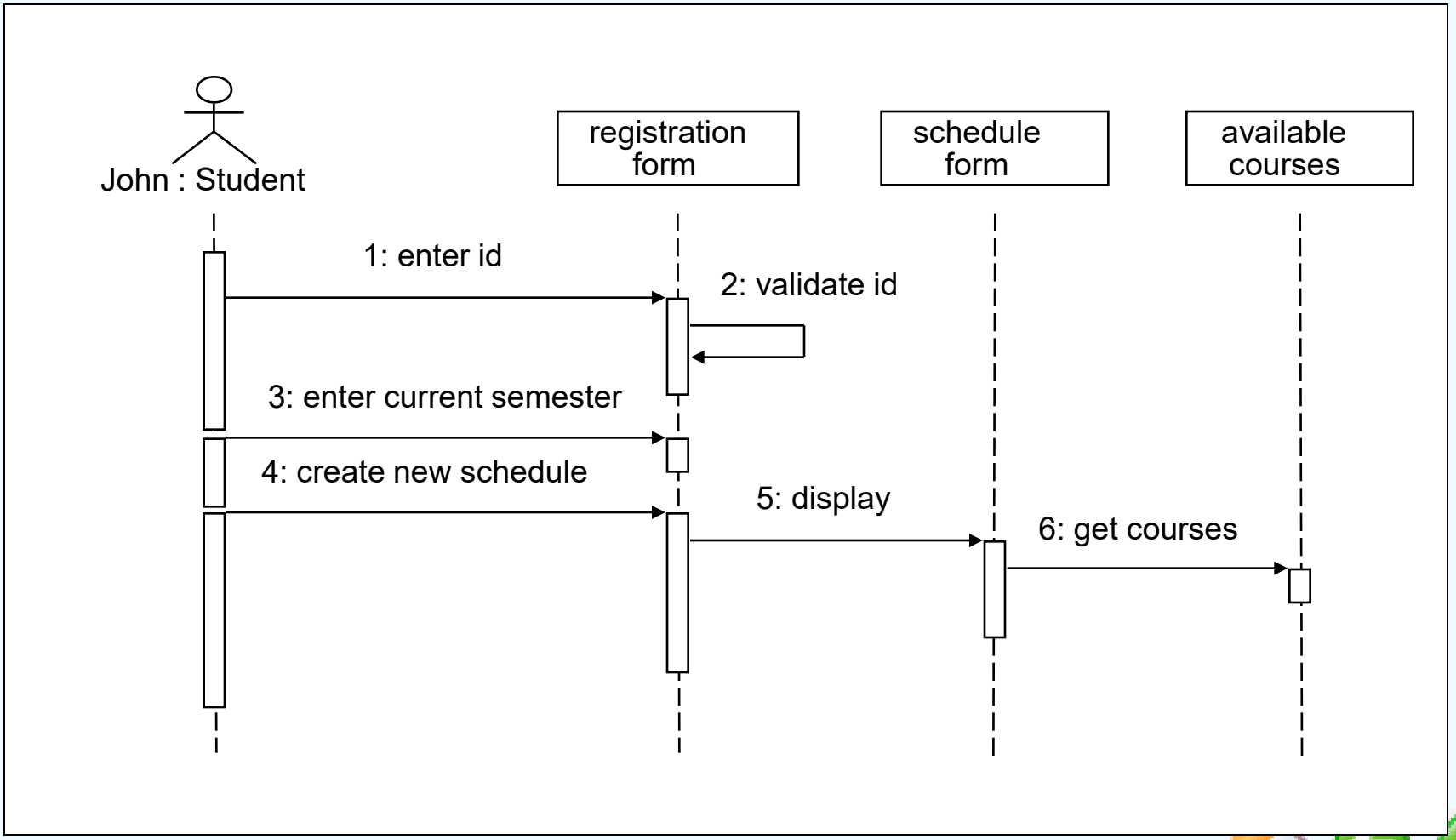


What is Focus of Control?

- Focus of Control represents the relative time that the flow of control is focused in an object
  - It represents the time an object is directing messages
- Focus of Control may be shown on a sequence diagram

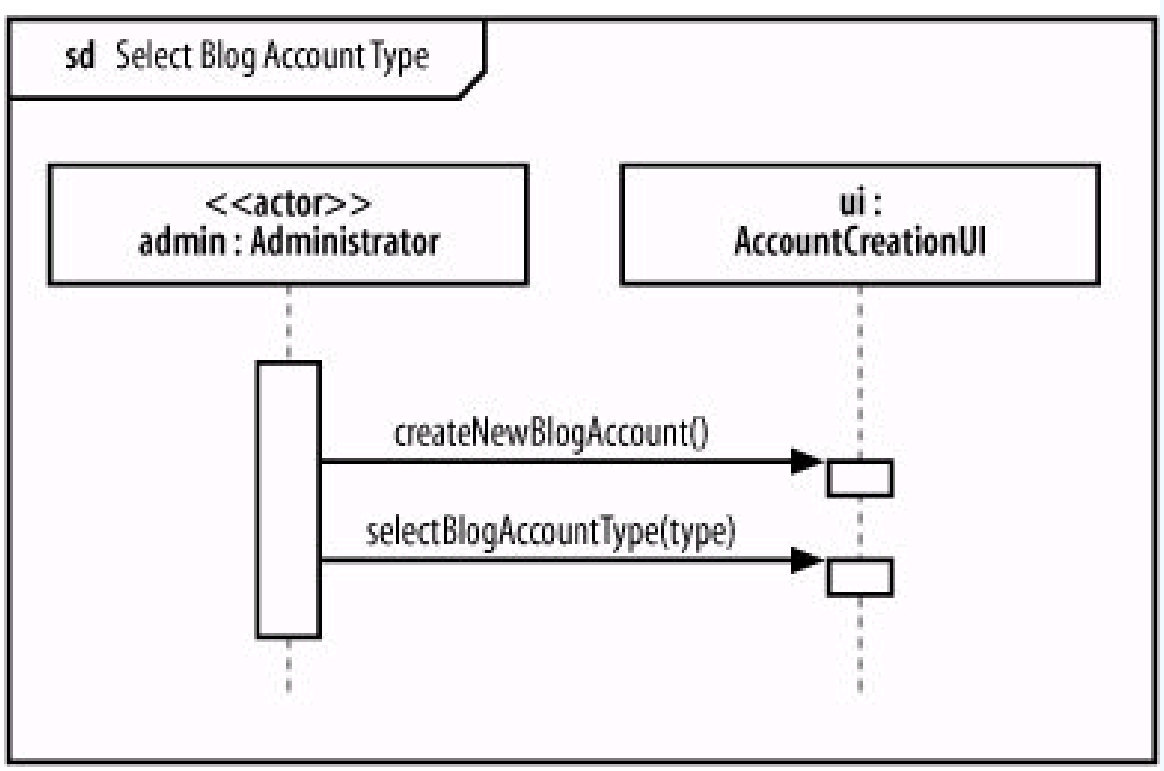


Focus of Control

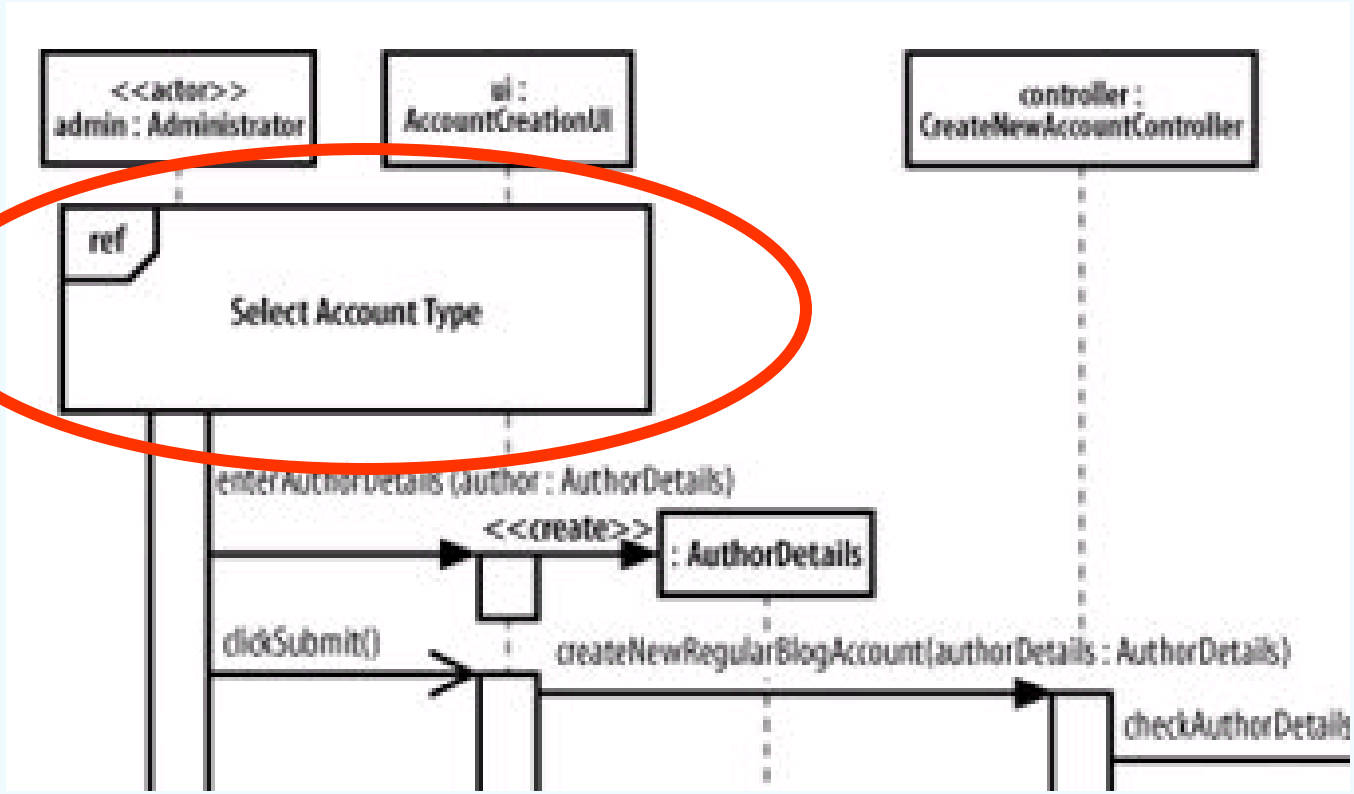


Fragments in sequence diagram

- **A sequence diagram may be broken into fragments**



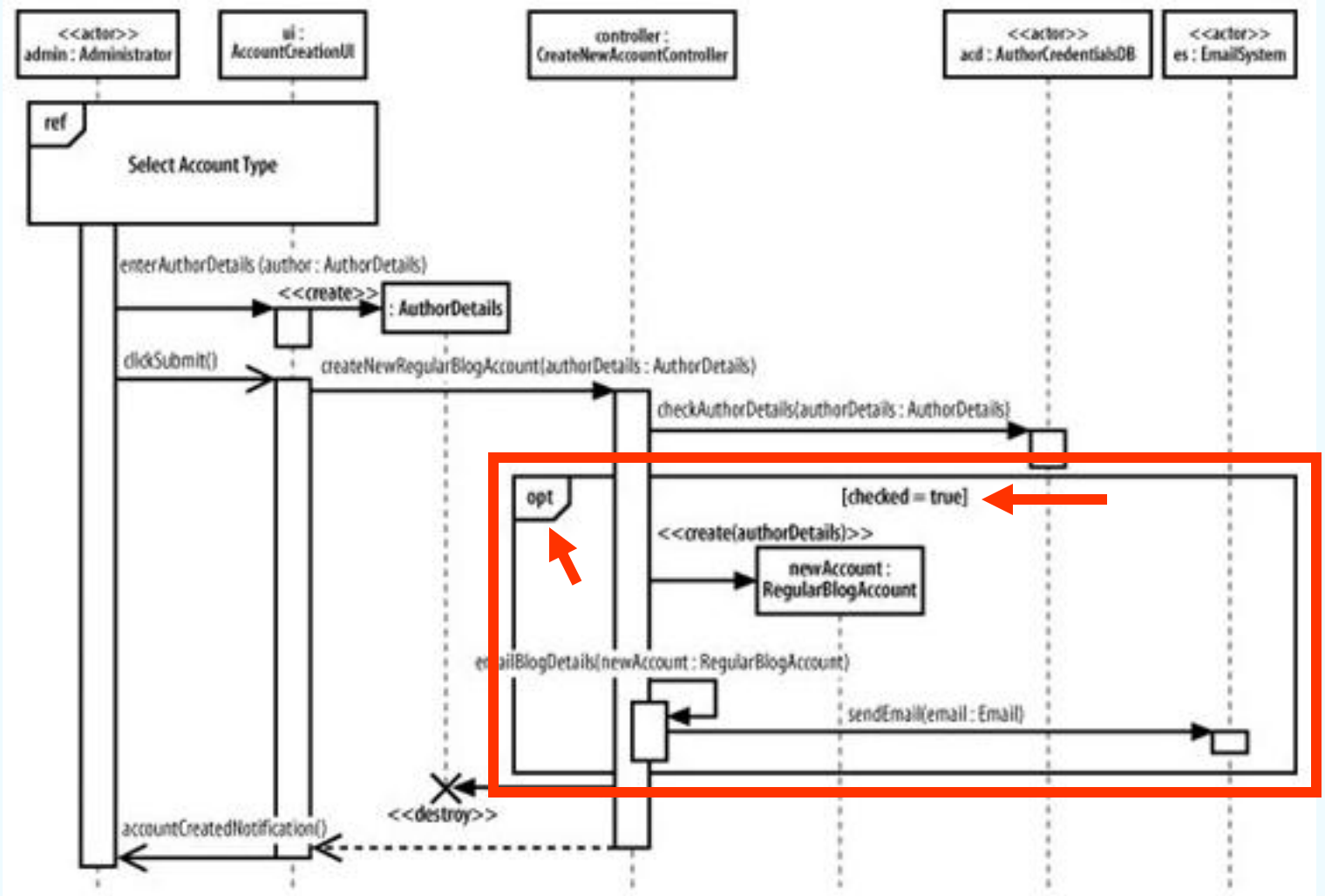
Using fragments



...

...

Using fragments (2)



Some fragment types

- ref: an interaction defined elsewhere
- loop: repeat the interactions in the fragment any times
- break: exit the enclosing loop fragment
- alt [guard conditions]: execute the corresponding set of interactions, base on which guard condition is true
- opt [check]: optional fragment, executed only if the check value is true



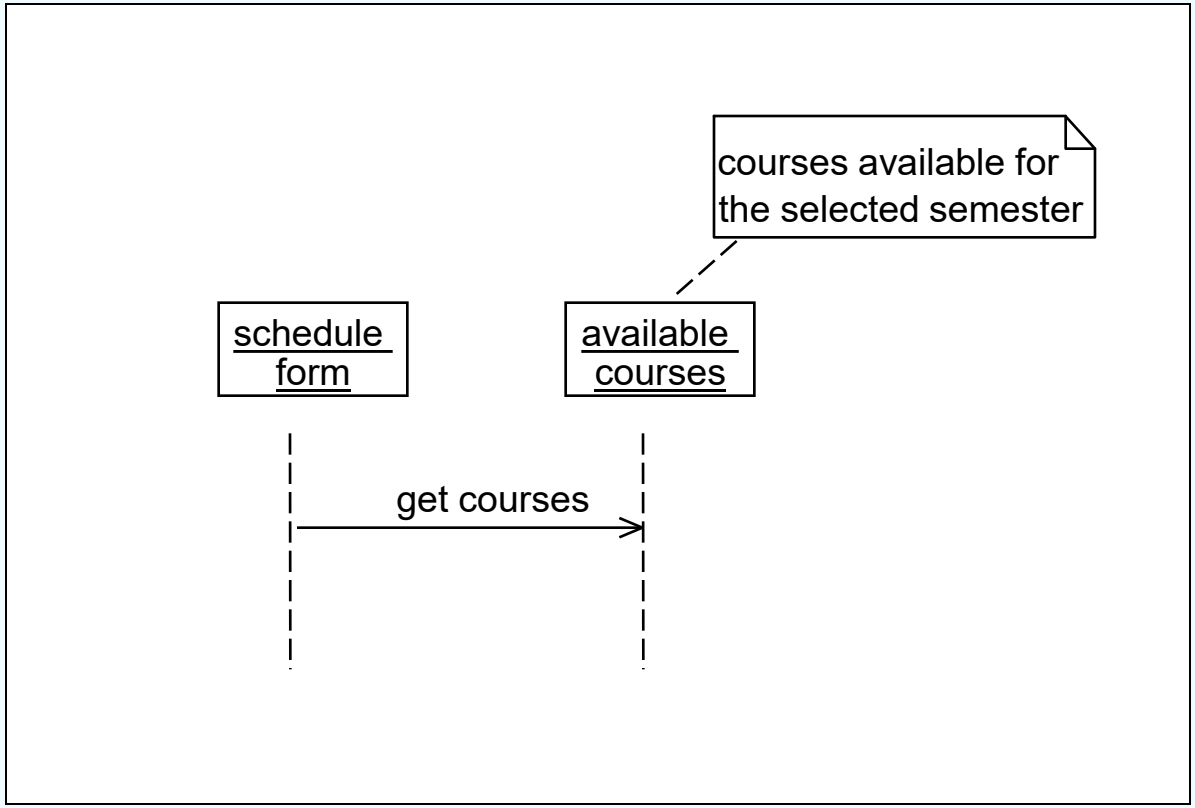
Some fragment types (2)

- par: the interactions can be executed in parallel
- region: the interactions in the fragment is said to be in a critical region (update shared paticipant)
- …



Notes

- Notes may be added to add more information to the diagram



Collaboration Diagrams (Communication Diagrams)

Collaboration Diagrams

- A collaboration diagram is an alternate way to represent the messages exchanged by a set of objects
- The diagram shows object interactions organized around the objects and their links to each other

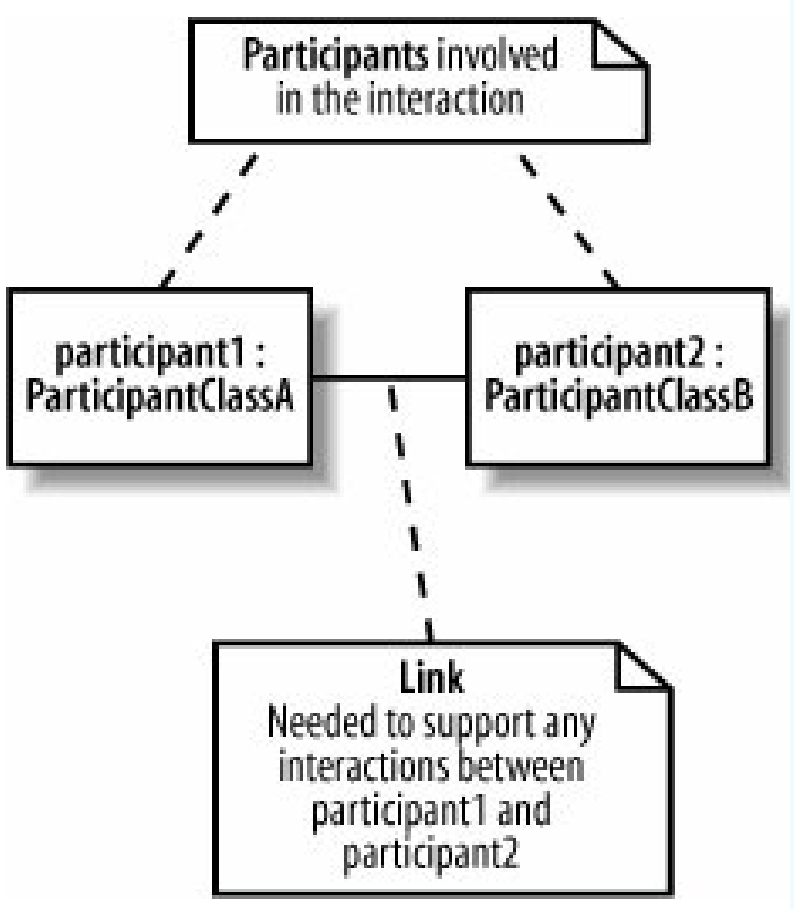


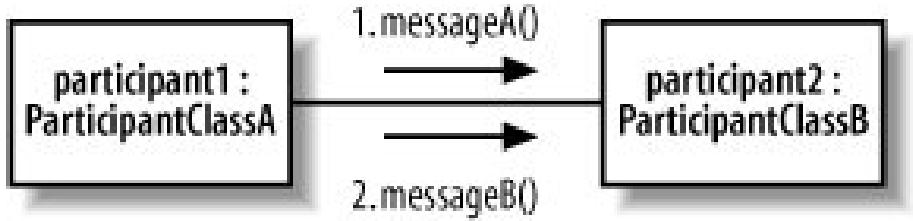
Collaboration Diagrams (2)

- **A collaboration diagram contains:**
  - Objects
  - Links between objects
  - Messages exchanged between objects
  - Data flowing between objects, if any



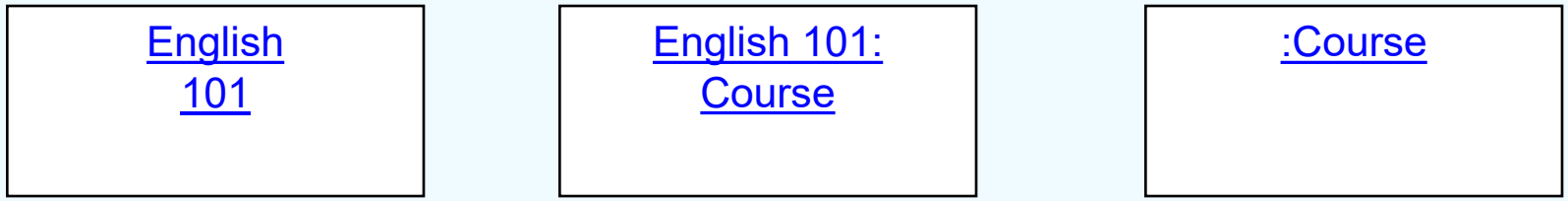
Participants, Messages and Links





Representing Objects

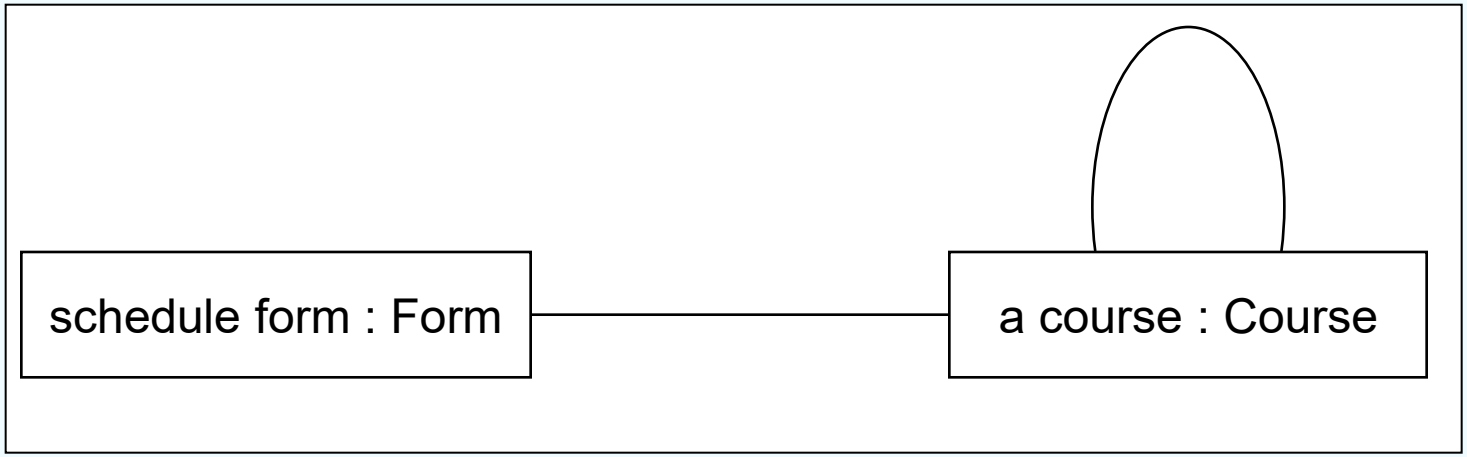
- Similar to that of the sequence diagram



Object only **Object and Class** **Class only**

Representing Links

- A interaction link in a collaboration diagram is represented as a line connecting object icons
- A link indicates that there is a pathway for communication between the connected objects

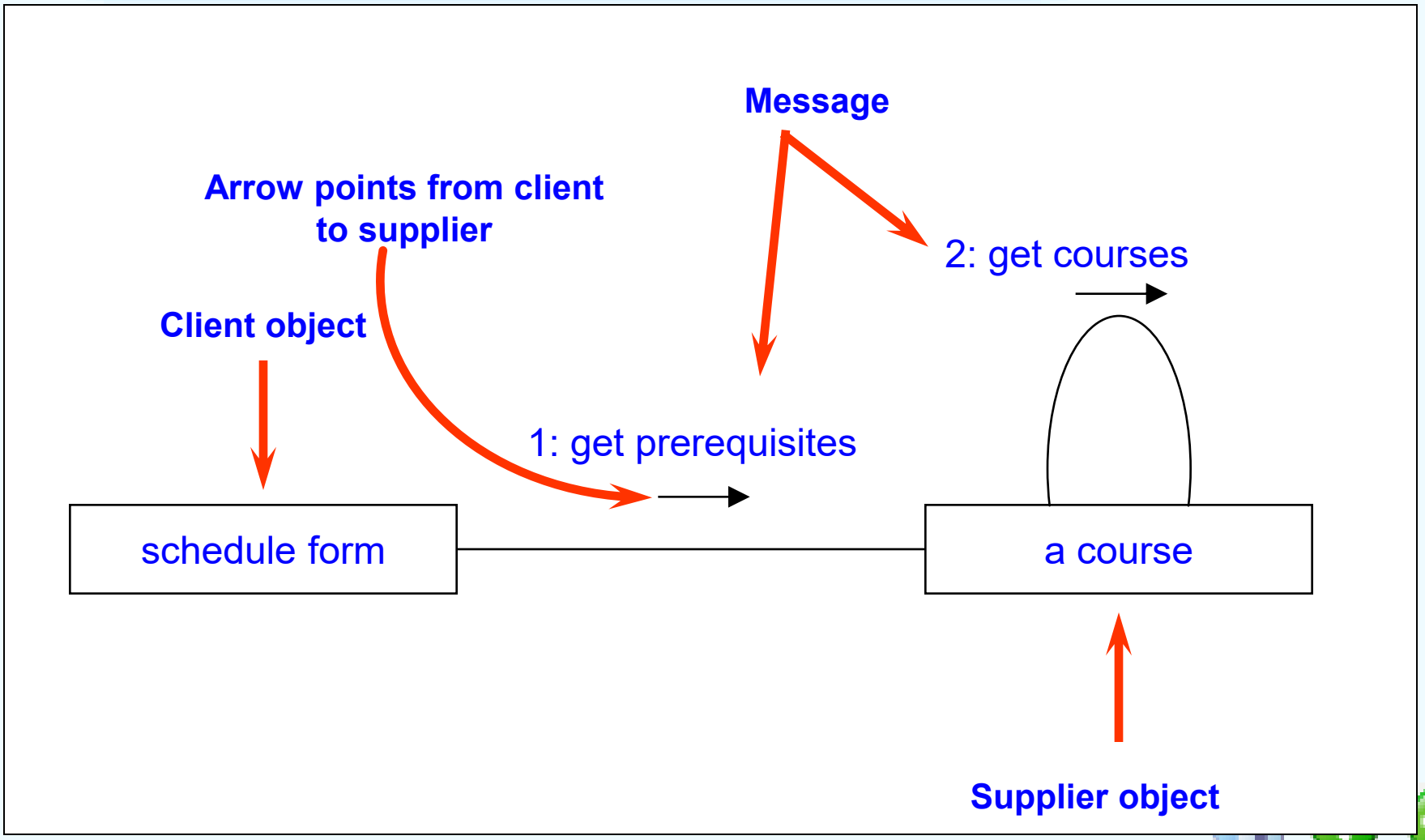


Link Annotations

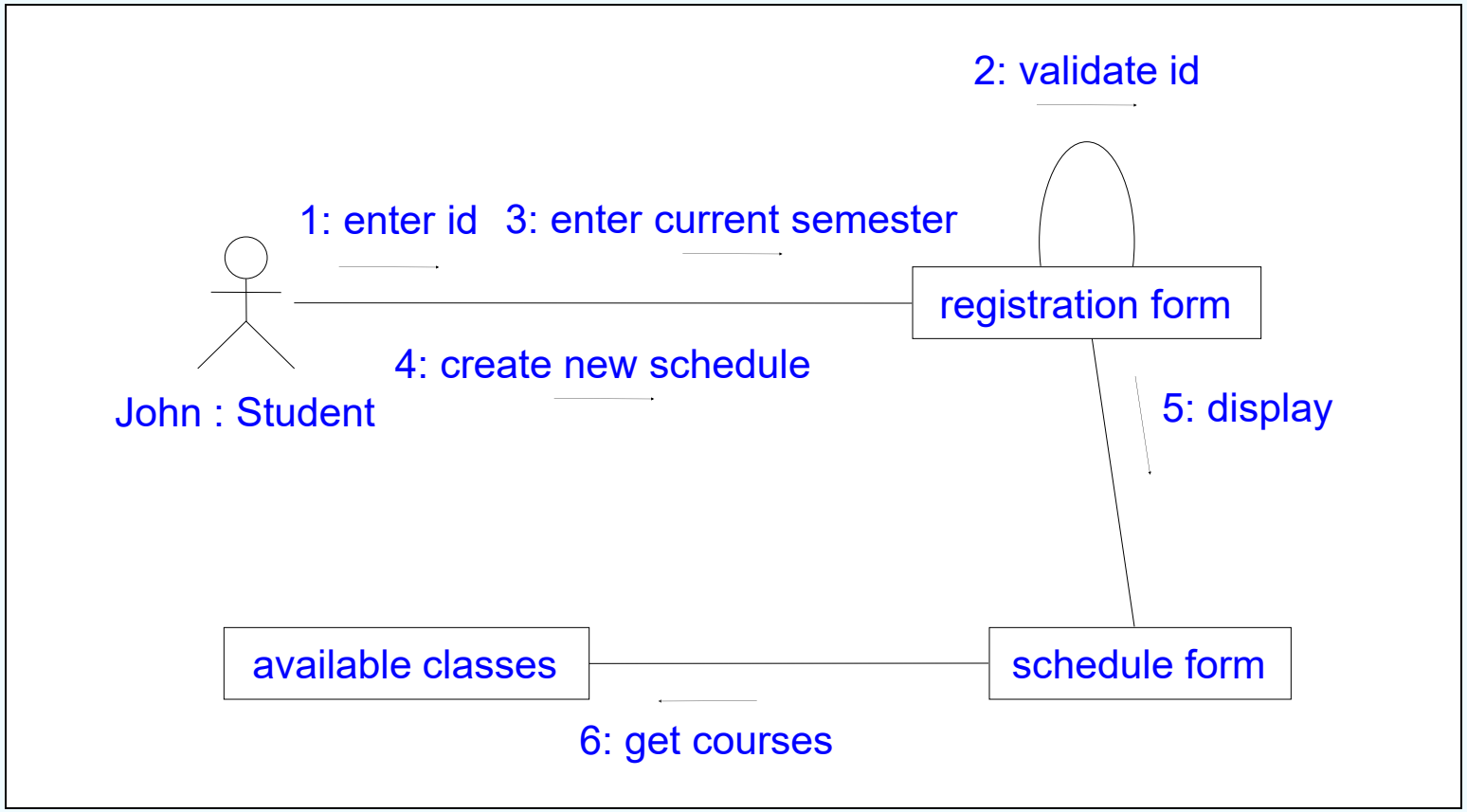
- An interaction link in a collaboration diagram can be annotated with:
  - An arrow pointing from the client object to the supplier object
  - The name of the message with an optional list of parameters and/or a data return value
  - An optional sequence number showing the relative order with which the messages are sent



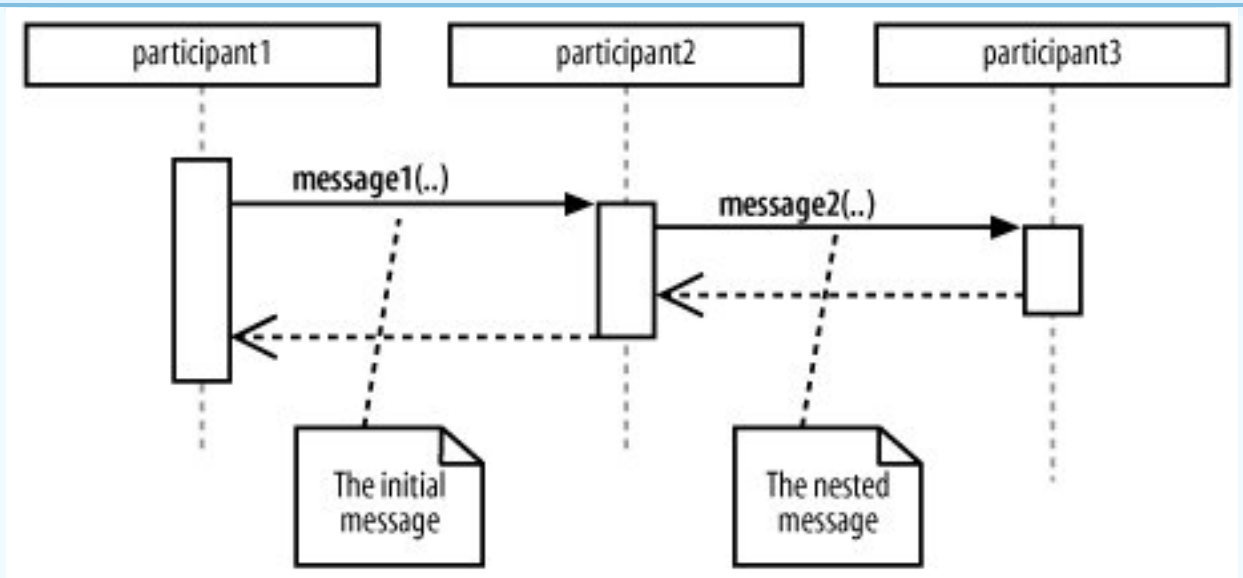
Link Notation

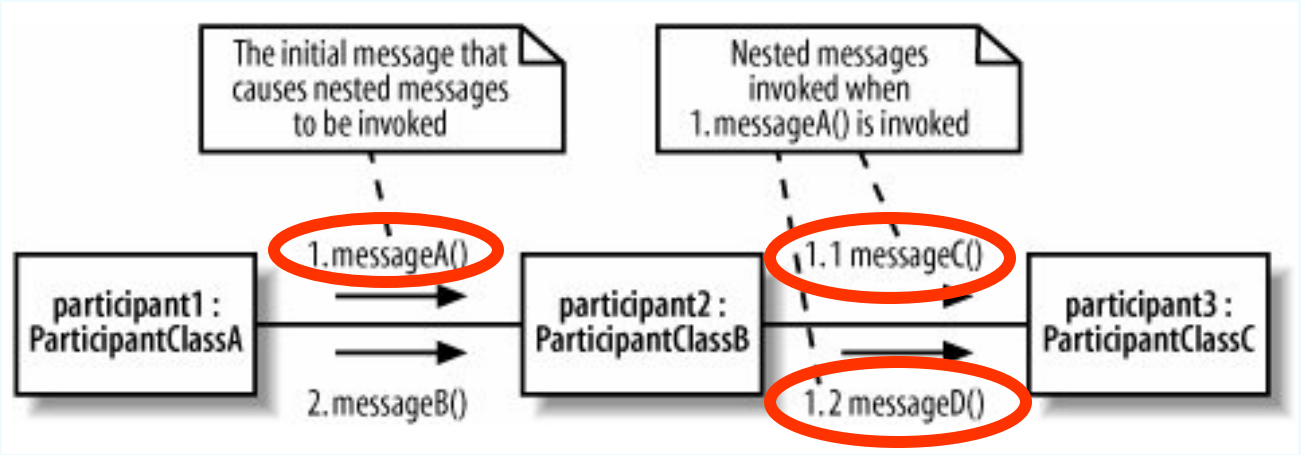


Sample Collaboration Diagram



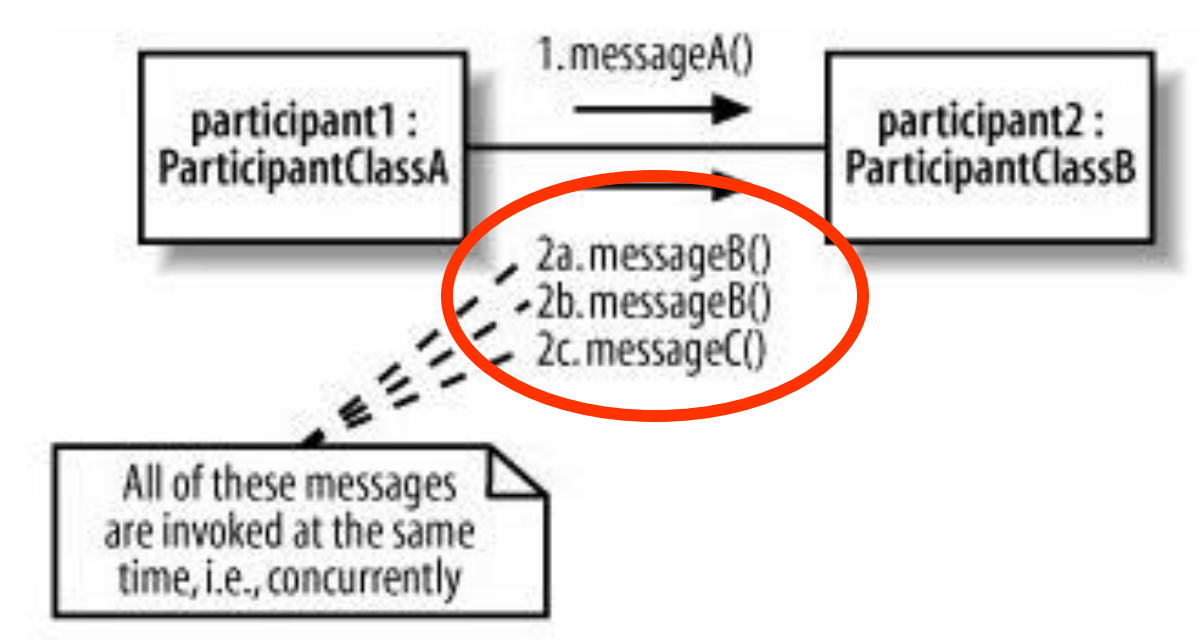
Nested messages





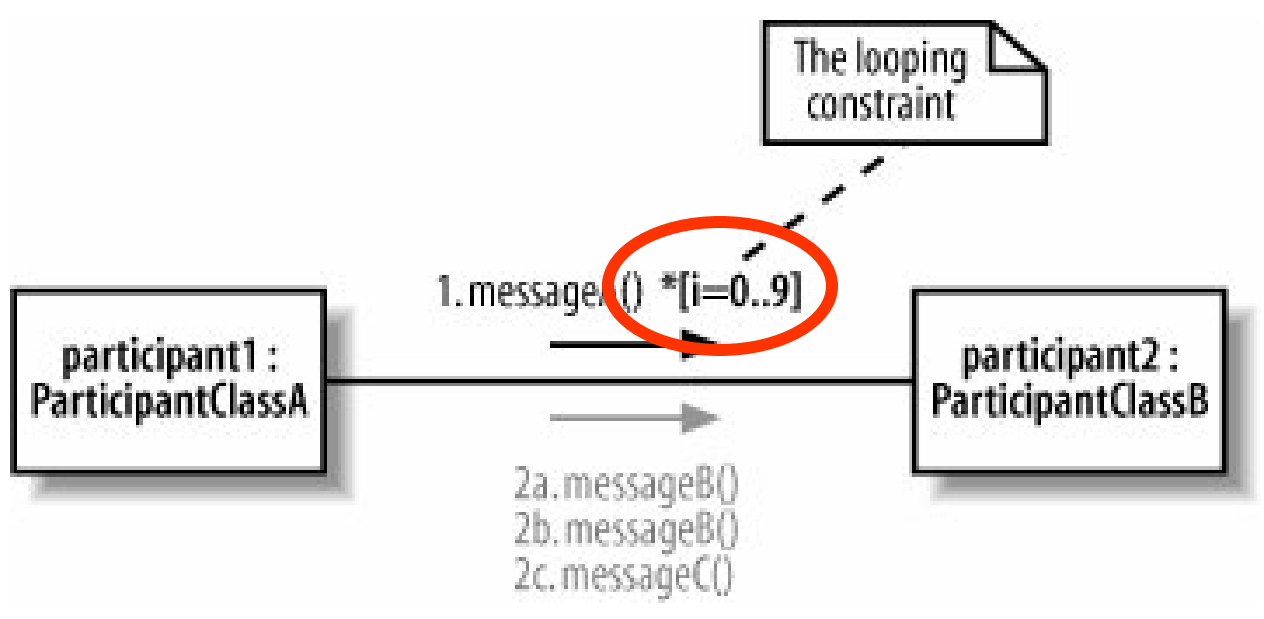
Concurrence messages

- **All concurrence messages have the same number, but different in the subsequent alphabetical character**



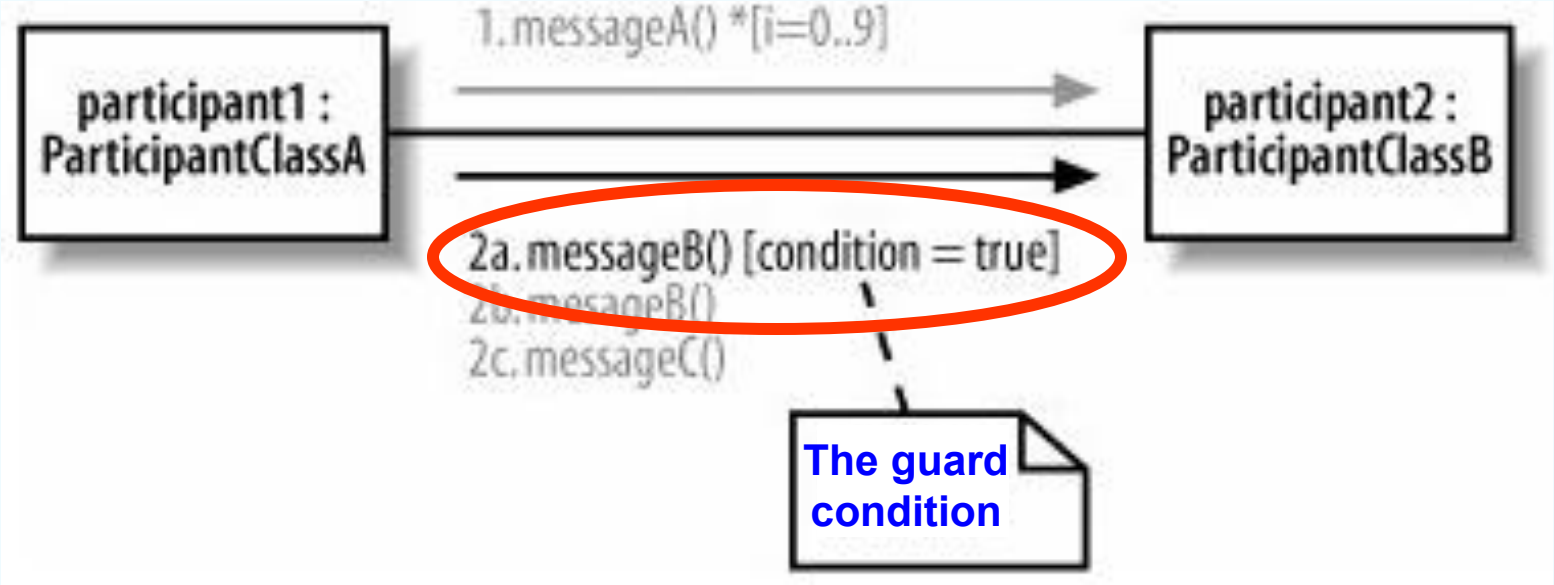
Messages invoked multiple times

- **Using multiplicity after the message**



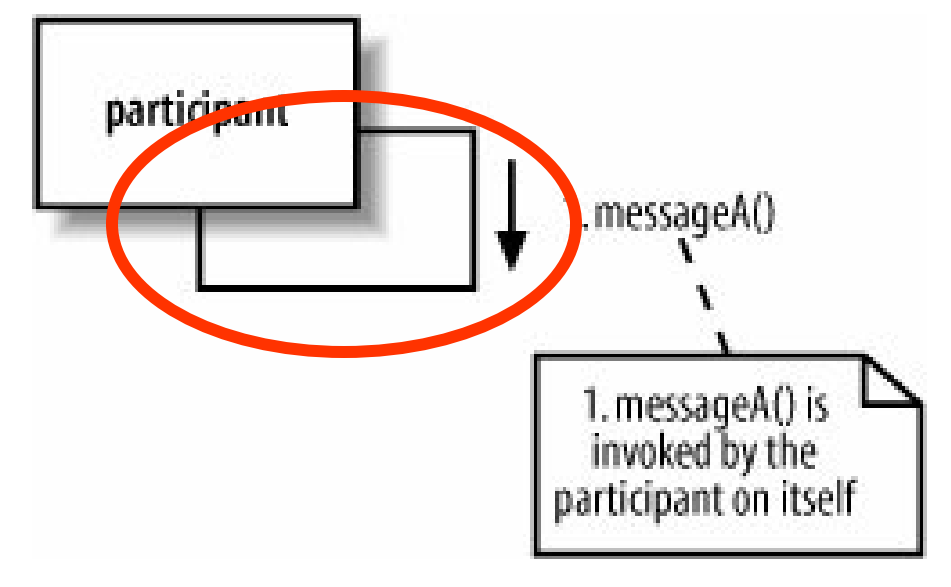
Messages based on a condition

- **The message can only be invoked if the condition is true**

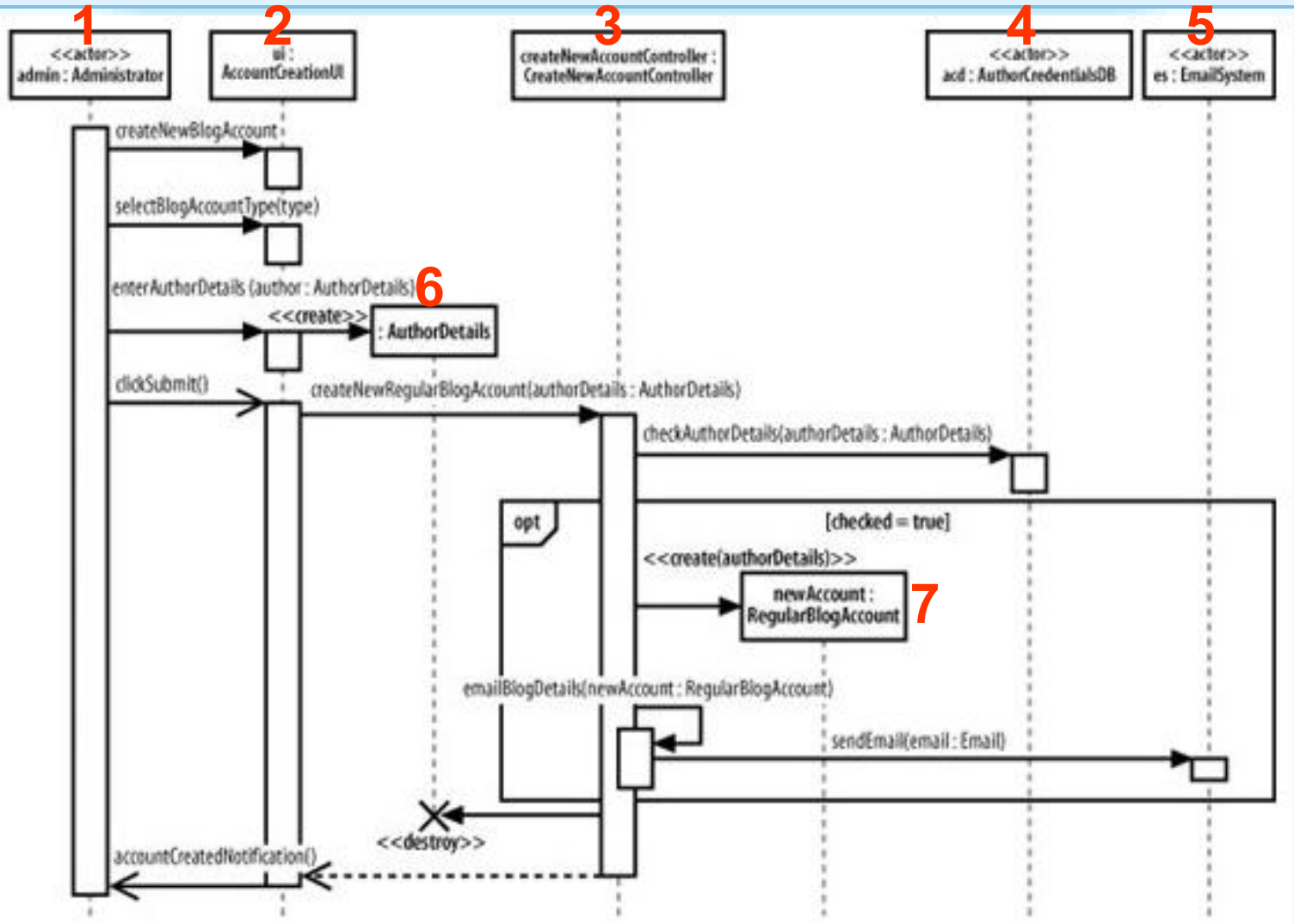


Self messages

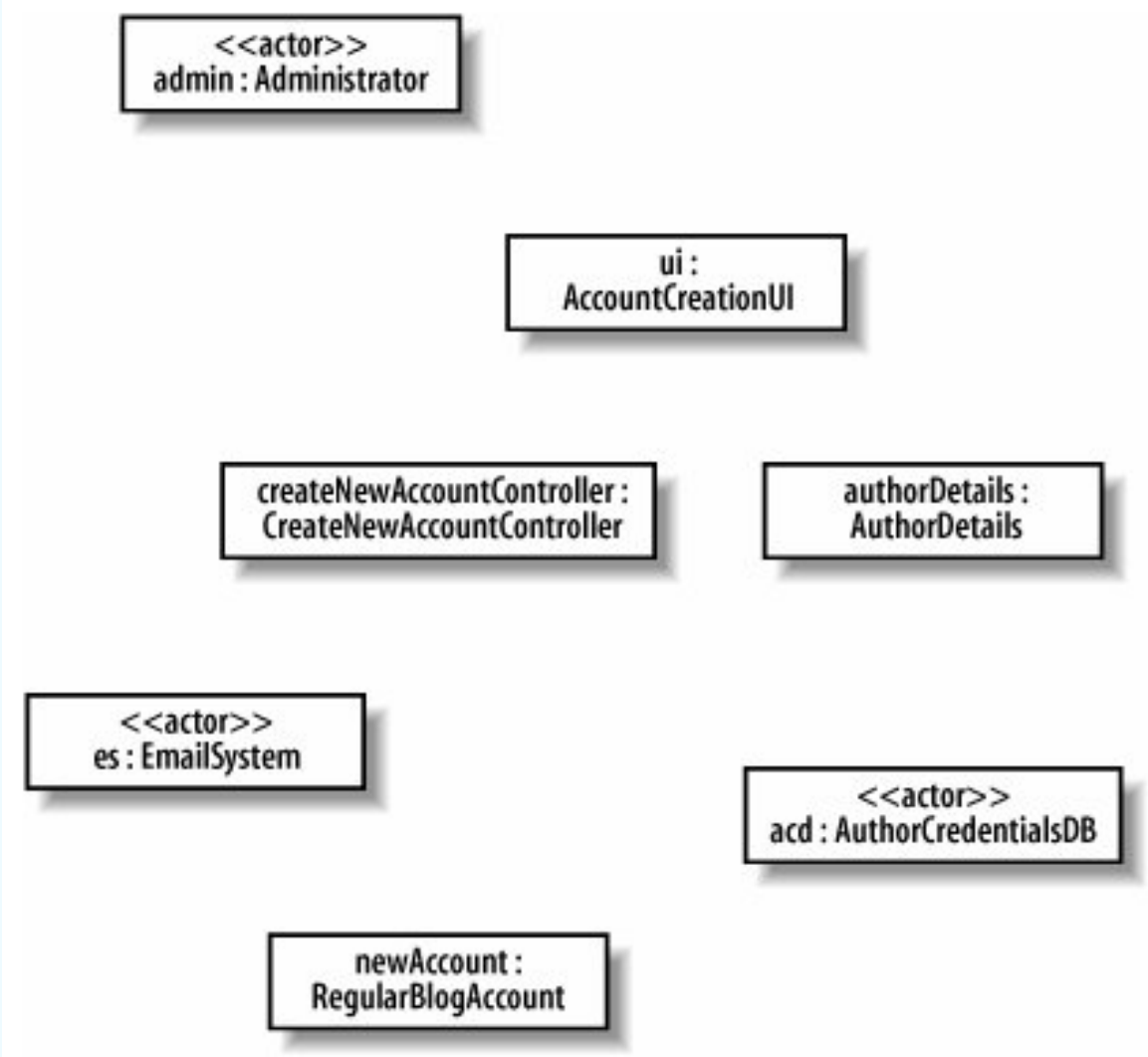
- **An object may send a message to itself**



Building a collaboration diagram

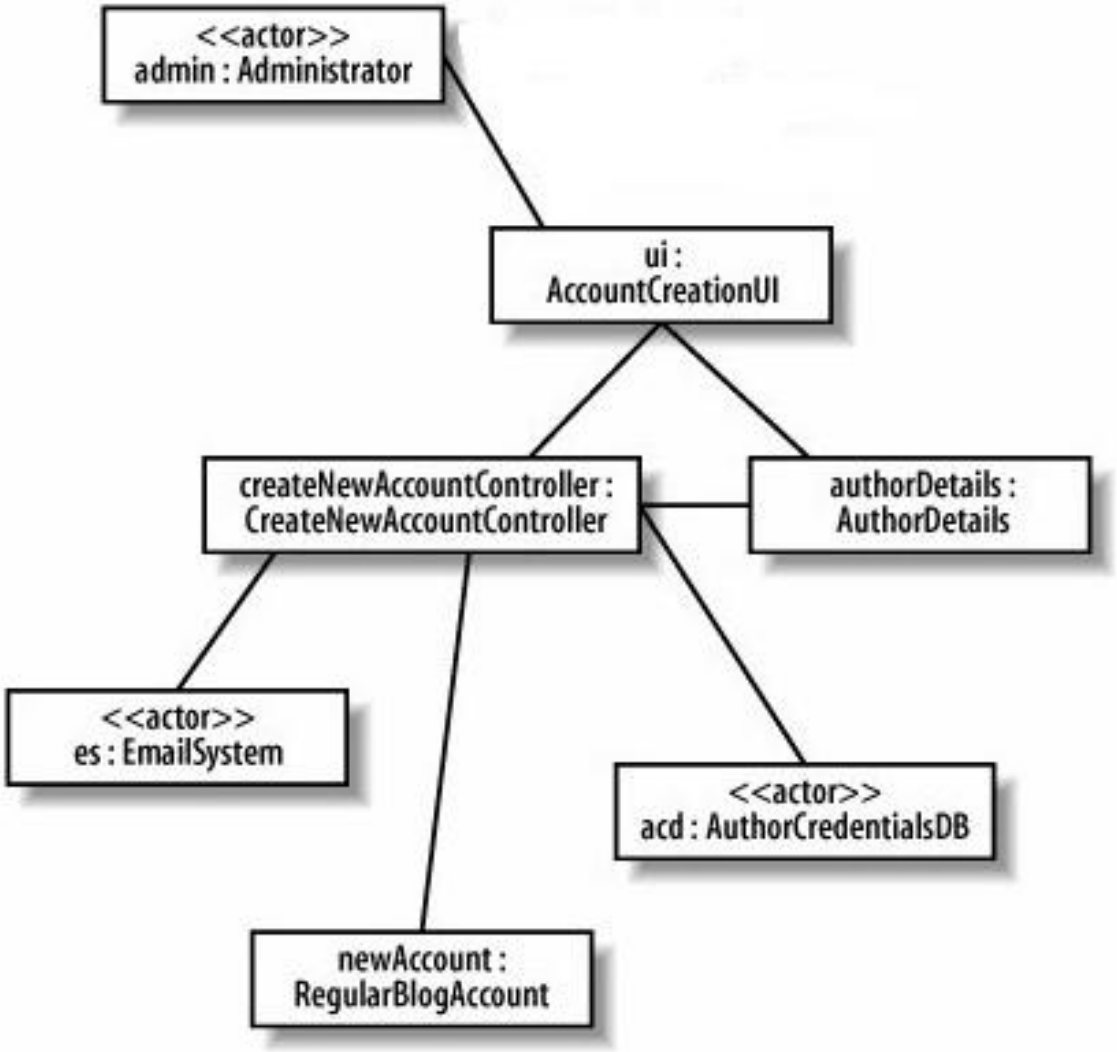


Building a collaboration diagram



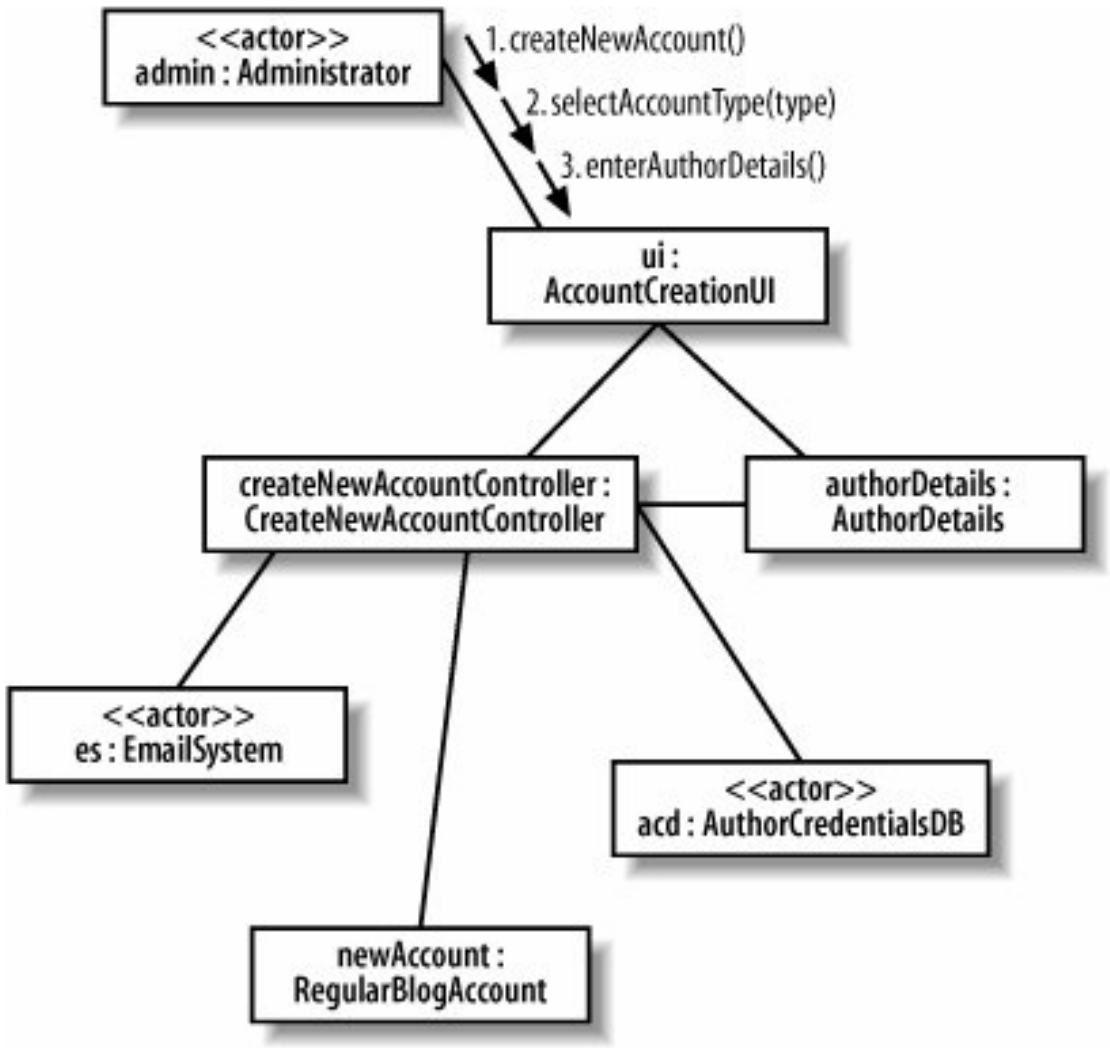
- Adding the participating objects into the communication diagram

Building a collaboration diagram



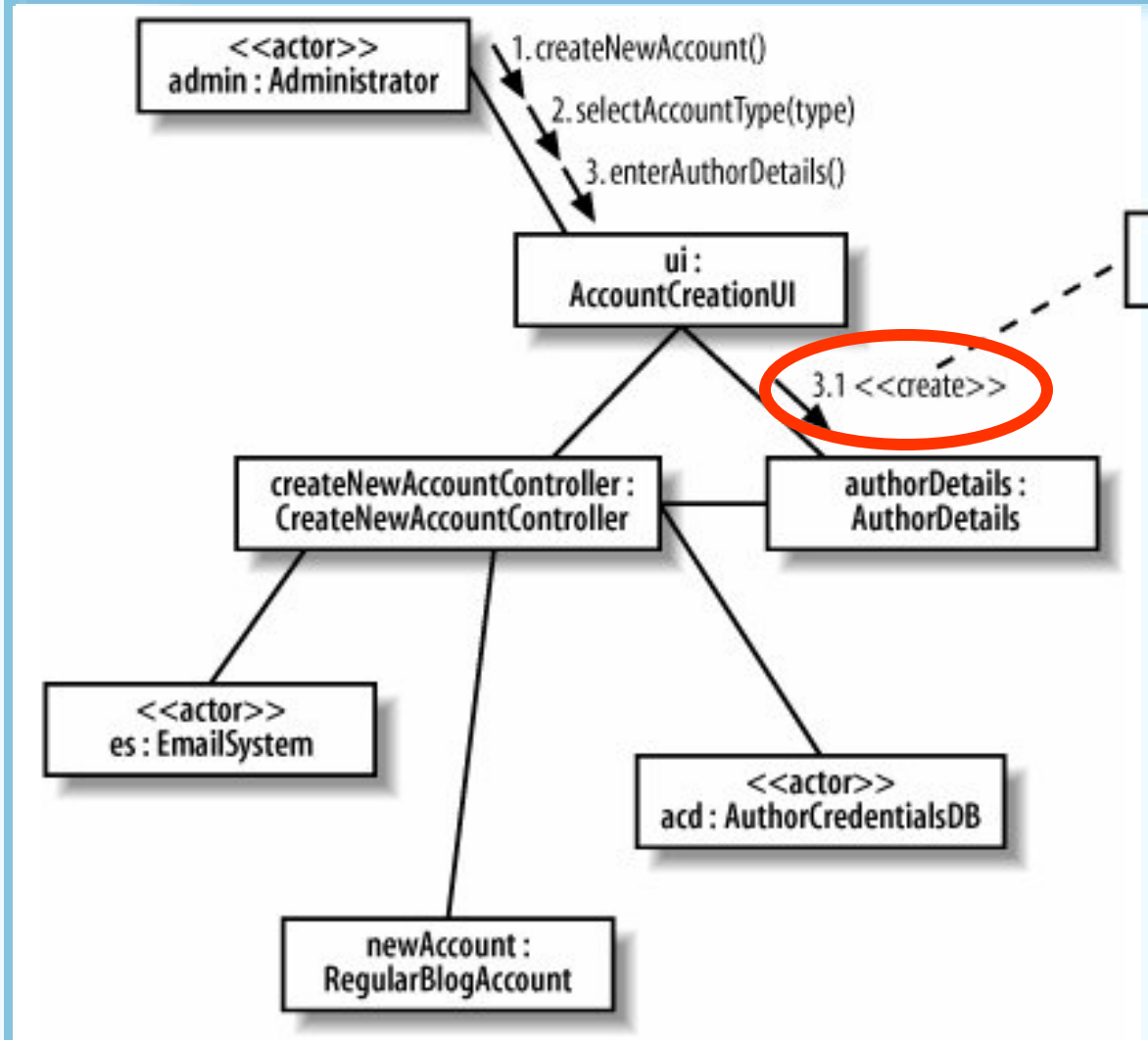
- Adding links required for the message passing to the communication diagram

Building a collaboration diagram



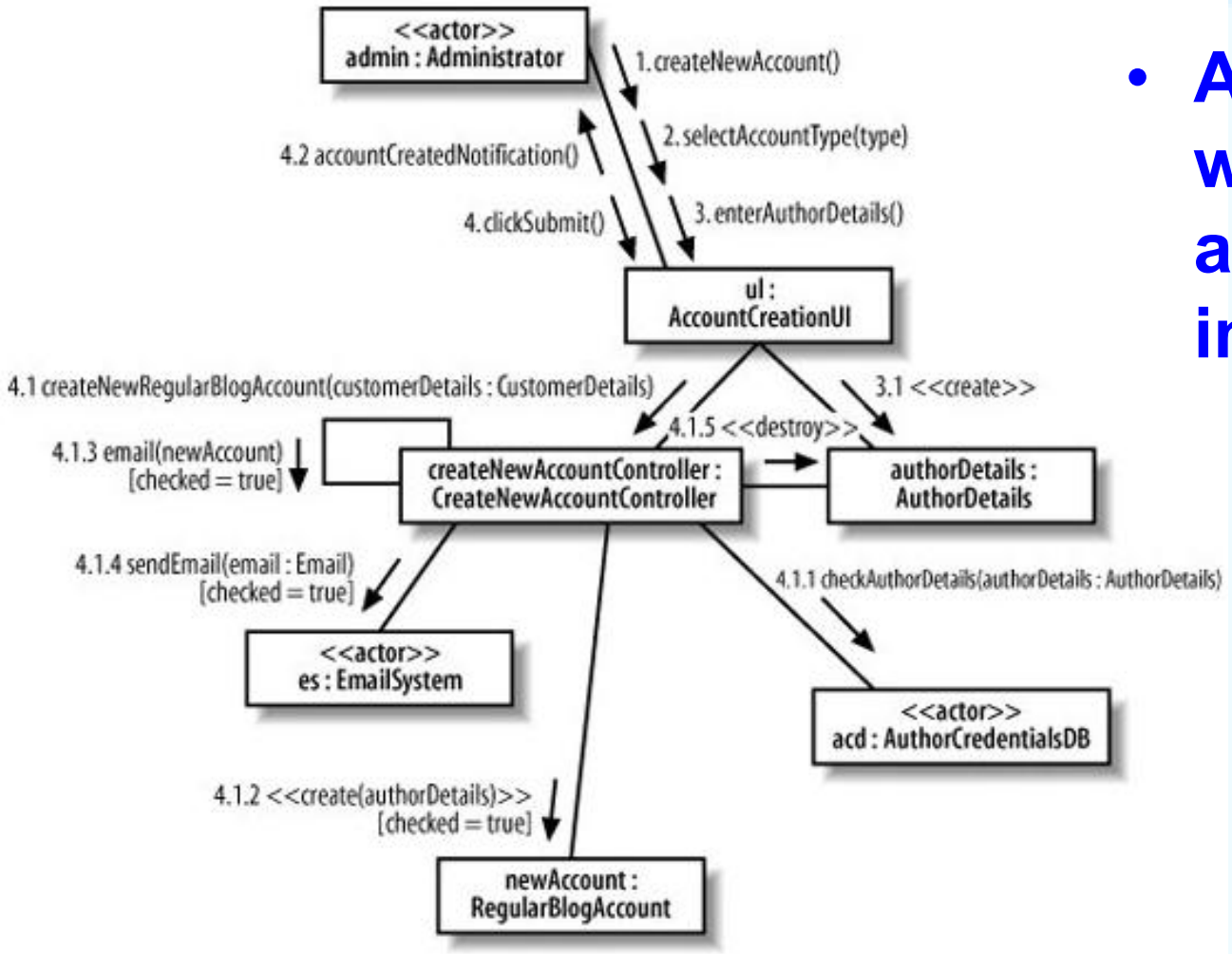
- 3 first separate messages are passed to the ui object, one after another

Building a collaboration diagram



- **The nested create message is invoked**

Building a collaboration diagram



- **And so on, until we finish adding all the messages into the diagram**

Summary: Object Interaction

- Object interaction can be represented graphically in a sequence diagram which shows the existence of objects and the interactions between the identified objects
  - Objects are represented by rectangles with underlined names
  - An object “lifeline” is represented by dashed vertical lines descending from the object



Summary: Object Interaction (2)

- Messages are indicated by horizontal arrows which are directed from the client object (sender) to the supplier object (receiver)
- The horizontal arrows are labeled with the message name
- An optional script may be added to provide more detail to the diagram



Summary: Object Interaction (3)

- A collaboration diagram is an alternate graphical representation of object interactions
  - Objects are represented by rectangles
  - An interaction link (line) is drawn between communicating objects
    - **The link is annotated with an arrow containing the message name which points from the client object to the supplier object**
    - **The link may also be annotated with a data return arrow**



Summary: Object Interaction (4)

- Sequence diagram vs Collaboration diagram
  - Show participants?
  - Show links?
  - Show message signatures?
  - Show parallel messages?
  - Show asynchronous messages?
  - Show message order?
  - Easy creation and maintaining?





THE END

